Solving Helmholtz Problems on Overset Grids by Time Filtering the Wave Equation

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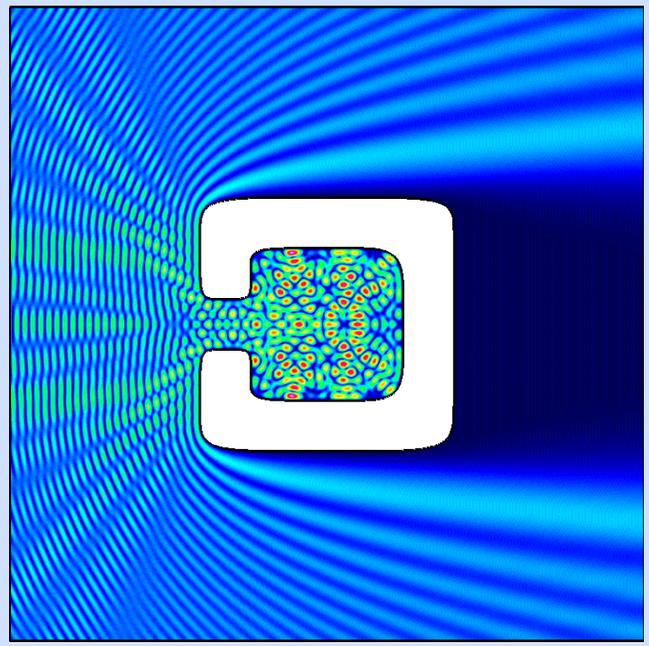
RPI



The 17th Symposium on Overset Composite Grids and Solution Technology, State College, Pennsylvania, USA, Sept 21-24, 2026.

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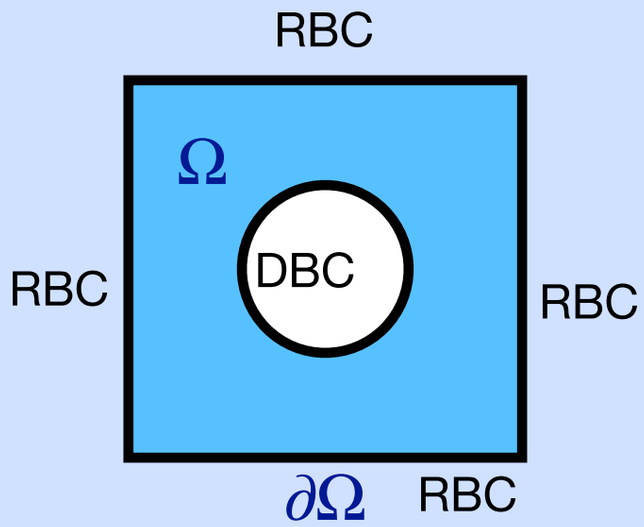
Goal: Find time periodic solutions to the forced wave equation on domains with closed and open boundaries

The wave equation (real valued)

$$
\begin{aligned}
&\partial_t^2 w = c^2 \Delta w - f(\mathbf{x}) \cos(\omega t), && \mathbf{x} \in \Omega, \\
&w(\mathbf{x}, 0) = w_0(\mathbf{x}), && \mathbf{x} \in \Omega \\
&\partial_t w(\mathbf{x}, 0) = w_1(\mathbf{x}), && \mathbf{x} \in \Omega, \\
&\mathcal{B}[\partial_t, \nabla] w = 0, && \mathbf{x} \in \partial\Omega,
\end{aligned}
$$



RBC = radiation boundary condition

Sample boundary conditions:

$$
\begin{aligned}
&\mathcal{B}[\partial_t, \nabla] w = \alpha_1 \partial_n w + \alpha_0 w, && \text{Dirichlet/Neumann/Mixed BC}, \\
&\mathcal{B}[\partial_t, \nabla] w = \partial_t w + c\gamma\, \partial_n w = 0, && \text{Radiation/impedance BC}
\end{aligned}
$$

The Helmholtz problem (complex valued)

$$
\begin{aligned}
&c^2 \Delta u + \omega^2 u = f(\mathbf{x}), && \mathbf{x} \in \Omega, \\
&\mathcal{B}[-i\omega, \nabla] u = 0, && \mathbf{x} \in \partial\Omega.
\end{aligned}
$$

Solving Helmholtz equations can be hard.

There are many ways to solve Helmholtz problems. For example:

1. Boundary integral, boundary element methods, fast multipole methods
2. Fast direct solvers
3. Domain decomposition with special transmission conditions, sweeping preconditioners, complex shifted Laplacian preconditioners, multigrid, ….

Possible issues with the current state of the art:

1. Poor performance for high frequencies.
2. Poor scaling (memory/CPU) as mesh is refined.
3. Pollution errors - low order methods require fine grids at high frequencies.
4. Poor parallel scaling or large initialization costs.
5. Limited to specialized geometries and or boundary conditions.

D.Lahaye, J.Tang, K.Vuik, Modern Solvers for Helmholtz Problems, Birkhauser, 2017

O.G. Ernst, M.J. Gander, Why it is Difficult to Solve Helmholtz Problems with Classical Iterative Methods, 2012.

Energy Conserving Problems (closed domains)

WaveHoltz Algorithm (closed domains, energy conserving case):

Goal: Solve the Helmholtz problem (e.g. with Dirichlet/Neumann BCs):

$$\begin{aligned} \mathcal{L}u + \omega^2 u &= f(\mathbf{x}), \quad && \mathbf{x} \in \Omega, \\ \mathcal{B}u &= g(\mathbf{x}), && \mathbf{x} \in \partial\Omega, \end{aligned}$$

$$\mathcal{L} = c^2 \Delta$$

Given an initial guess $v^{(k)} \approx u(x)$ solve the forced wave equation:

$$\begin{aligned} \partial_t^2 w &= \mathcal{L}w - f(\mathbf{x})\cos(\omega t), && \mathbf{x} \in \Omega,\ t \in [0, \bar{T}] \\ \mathcal{B}w &= g(\mathbf{x})\cos(\omega t), && \mathbf{x} \in \partial\Omega, \\ w(\mathbf{x}, 0) &= v^{(k)}, && \mathbf{x} \in \Omega, \\ \partial_t w(\mathbf{x}, 0) &= 0, && \mathbf{x} \in \Omega. \end{aligned}$$

$$\bar{T} = N_p \frac{2\pi}{\omega}$$

Number of periods to integrate over

Current iterate is the initial condition

Compute $v^{(k+1)}$ from the time filter of w:

$$v^{(k+1)}(\mathbf{x}) = \frac{2}{\bar{T}} \int_0^{\bar{T}} \left(\cos(\omega t) - \frac{\alpha}{2} \right) w^{(k)}(\mathbf{x}, t; v^{(k)})\, dt,$$

New iterate is the time filtered solution to the wave equation

WaveHoltz transformed the problem to one that is easier to solve

The WaveHoltz algorithm defines an affine operator:

$$v^{(k+1)}(\mathbf{x}) = \mathcal{A} v^{(k)} + b$$

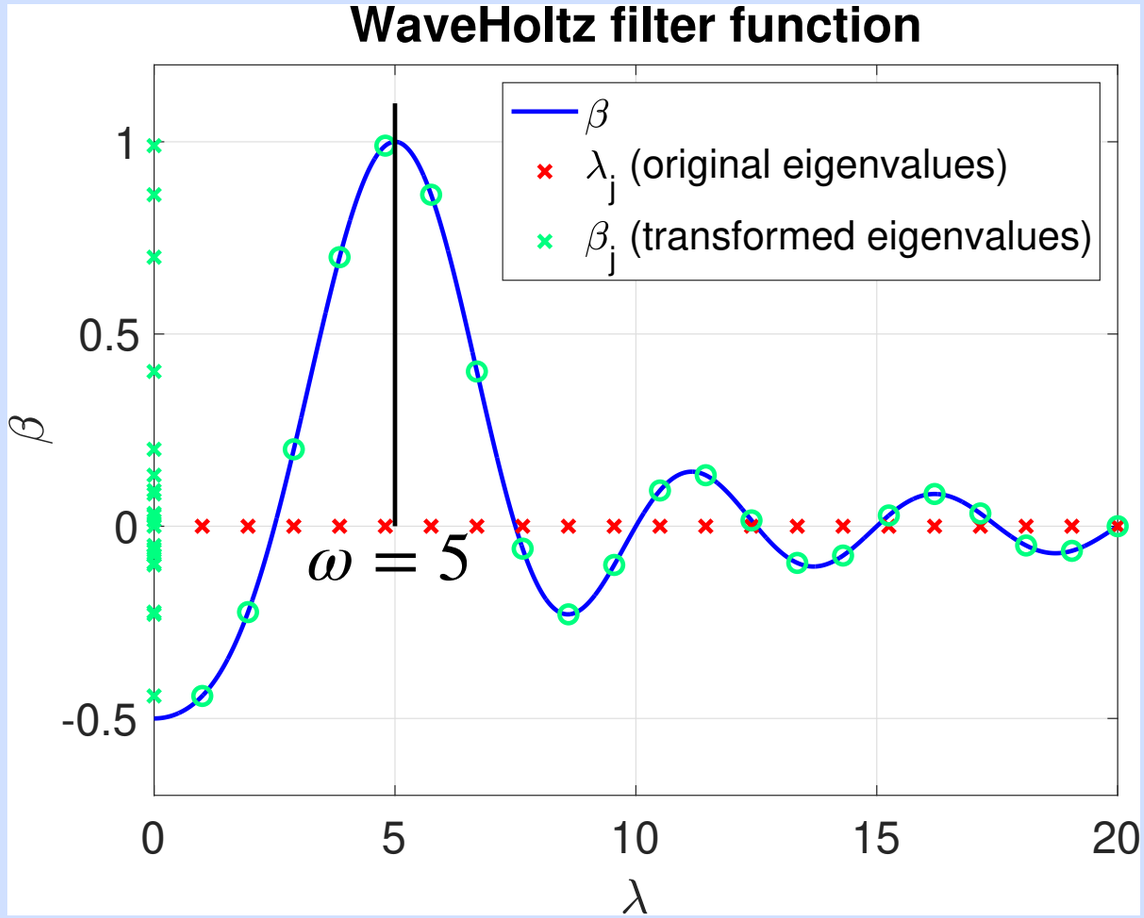
Where $\mathcal{A}$ has the same eigenfunctions as the Helmholtz (or Δ) operator:

$$\mathscr{L}\phi_j = -\lambda_j^2 \phi_j \longrightarrow \mathscr{A}\phi_j = \beta_j \phi_j$$

But eigenvalues:

$$\beta_j \overset{\text{def}}{=} \beta(\lambda_j, \omega)$$

$$\beta(\lambda;\omega) = \frac{2}{\bar{T}} \int_0^{\bar{T}} \left(\cos(\omega t) - \frac{1}{4}\right) \cos(\lambda t)\, dt$$



The transformed problem has eigenvalues $\beta_j \in [-.5, 1]$, largest near ω, and more easily solved with traditional iterative solvers! No need to invert an indefinite matrix.

Enhancements to the basic WaveHoltz algorithm:

1. Implementation on overset grids (here we use the Overture codes CgWave and CgWaveHoltz)
2. Acceleration by matrix-free Krylov methods such as GMRES
3. Implicit time-stepping with a large time-step
4. Deflation using precomputed eigenvalues and eigenvectors

Eigenvector deflation for energy conserving problems

1. The WaveHoltz iteration can be accelerated using eigenvector deflation
2. The eigenvalues/eigenvectors can be computed using EigenWave, which is based on the WaveHoltz algorithm, or by some other method.
3. The slowest converging eigenvectors are chosen for deflation.

We consider two approaches to deflation

1. Direct Eigenvector Deflation (DEV)
2. Augmented Krylov Eigenvector Deflation (AUKED)

D.A. Appelo, WDH, Z. Peng, Numerical study of Eigenvector Deflation to Accelerate the WaveHoltz Method, ArXiv, 2026.

"Direct" eigenvector deflation accelerates convergence

1. Uses precomputed eigenvalues λ_m and eigenvectors ϕ_m
2. Typically deflate eigenpairs with $\lambda_m \approx \omega$

Algorithm 1 WaveHoltz Algorithm with Eigenfunction Deflation.

function WAVEHOLTZ(ω, f, N_p)
  // Final time is $\bar{T} = N_p T$ where $T = 2\pi/\omega$.
  $k = 0$ ▷ WaveHoltz iteration counter.
  $v^{(k)} = 0$ ▷ Initial guess for Helmholtz iterate (deflate if non-zero)
  $f_d = f - \sum_{\phi_m \in \mathcal{D}} (f, \phi_m)_\Omega \, \phi_m$ ▷ Deflate forcing.
  while not converged **do** ▷ Start WaveHoltz iterations.
    $w^{(k)}(\mathbf{x}, 0) = v^{(k)}(\mathbf{x})$ ▷ Initial condition for wave equation solve.
    $w^{(k)}(\mathbf{x}, 0 : \bar{T}) =$ SOLVEWAVEEQUATION$(w^{(k)}(\mathbf{x}, 0), f_d)$ ▷ Solve for $\mathbf{w}(\mathbf{x}, t)$ (deflated forcing f_d).
    $v^{(k+1)}(\mathbf{x}) = \frac{2}{\bar{T}} \int_0^{\bar{T}} \left(\cos(\omega t) - \frac{\alpha}{2} \right) w^{(k)}(\mathbf{x}, t) \, dt$ ▷ Time filter.
    $v^{(k+1)} = v^{(k+1)} - \sum_{\phi_m \in \mathcal{D}} (v^{(k+1)}, \phi_m)_\Omega \, \phi_m$ ▷ Deflate iterate (skip with true eigenfunctions).
    $k = k + 1$
  end while ▷ End WaveHoltz iterations.
  $v^{(k)} = v^{(k)} + \sum_{\phi_m \in \mathcal{D}} \frac{(f, \phi_m)_\Omega}{\omega^2 - \lambda_m^2} \phi_m$ ▷ Inflate.
  $u = v^{(k)}$; ▷ Approximate Helmholtz solution.
end function

(1) Deflate forcing

(2) Inflate solution

On overset grids line 10 is added since eigenvectors are only approximately orthogonal

Augmented Krylov Eigenvector Deflation (AUKED)

Krylov methods (e.g. GMRES) to solve $Ax = b$ look for solutions x in the Krylov space

$$x \in K_j = \{b, Ab, A^2b, \ldots, A^{j-1}b\}$$

Augmented Krylov methods look for solutions x in the augmented Krylov space

$$x \in \{w_1, w_2, \ldots, w_{N_a}\} \cup K_j$$

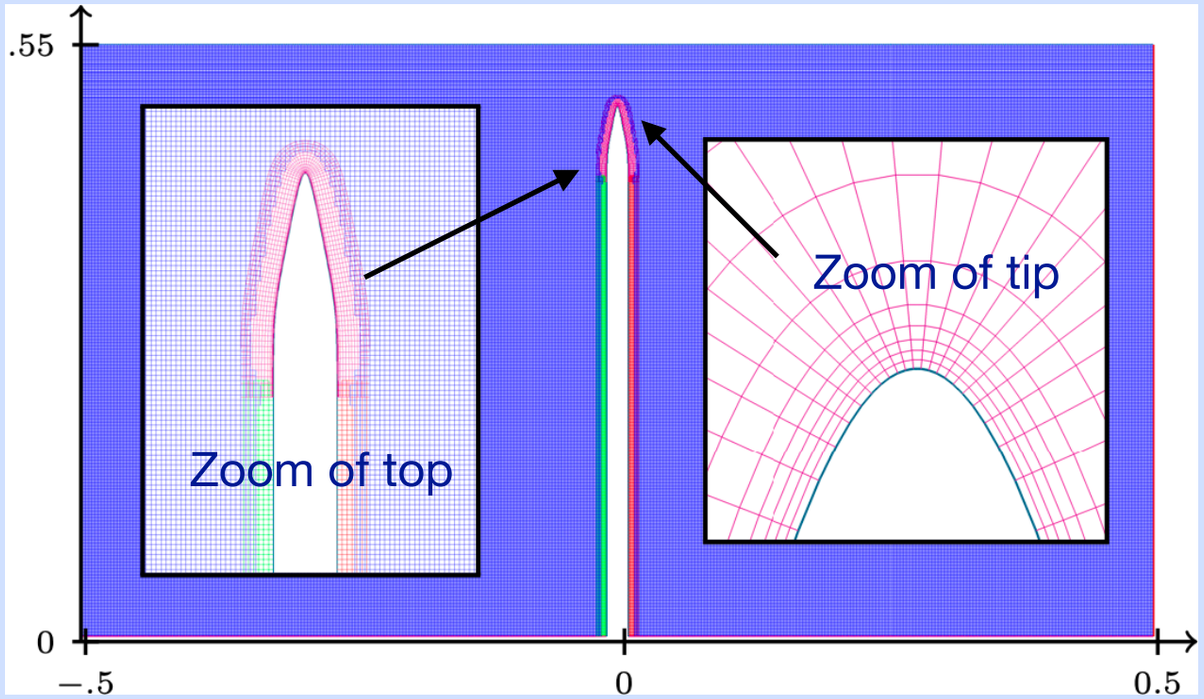
The augmented vectors w_k can be eigenvectors or approximate eigenvectors

Papers describing augmented GMRES and recycled bi-CG-Stab:

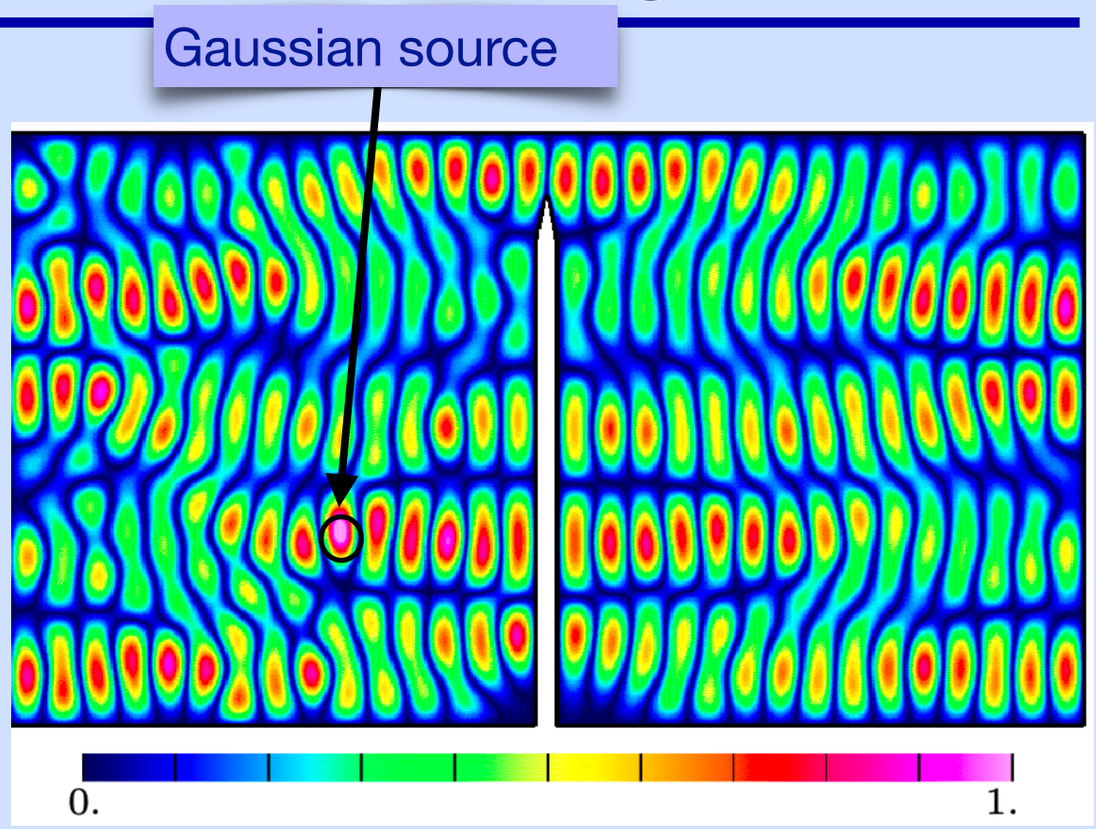
J. Baglama, L. Reichel, Augmented GMRES-type methods, Numerical Linear Algebra with Applications 14 (4) (2007) 337–350.

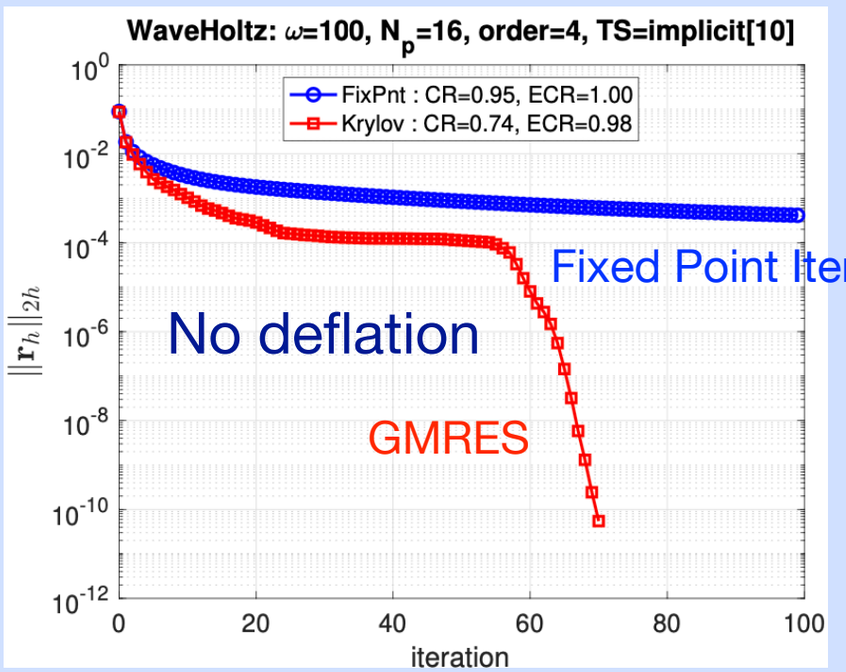
A. Amritkar, E. de Sturler, K. Świrydowicz, D. Tafti, K. Ahuja, Recycling Krylov subspaces for CFD applications and a new hybrid recycling solver, Journal of Computational Physics 303 (2015) 222–237.

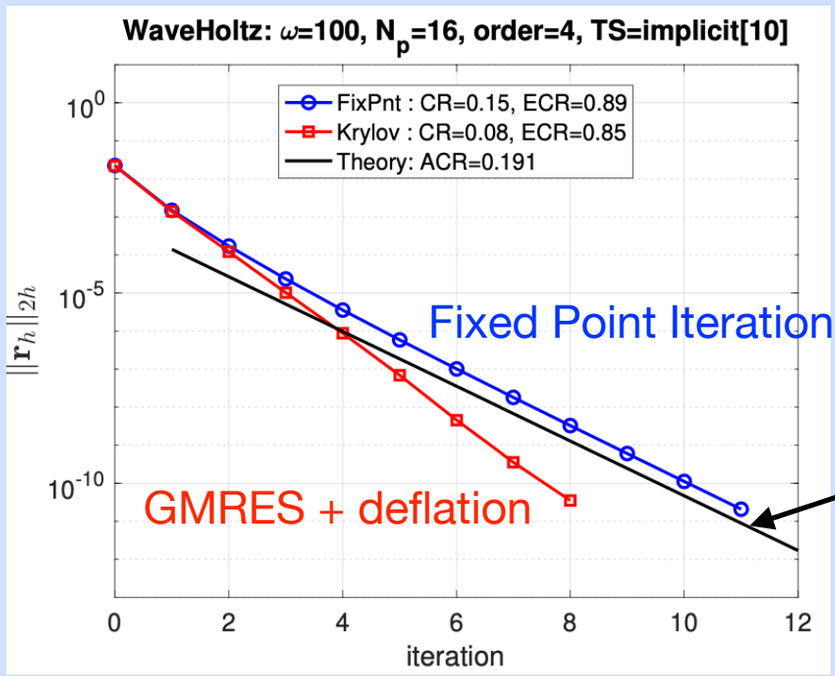
Comparison: Direct versus no deflation for a knife edge, $\omega = 100$



Small cells pose no problem with implicit method



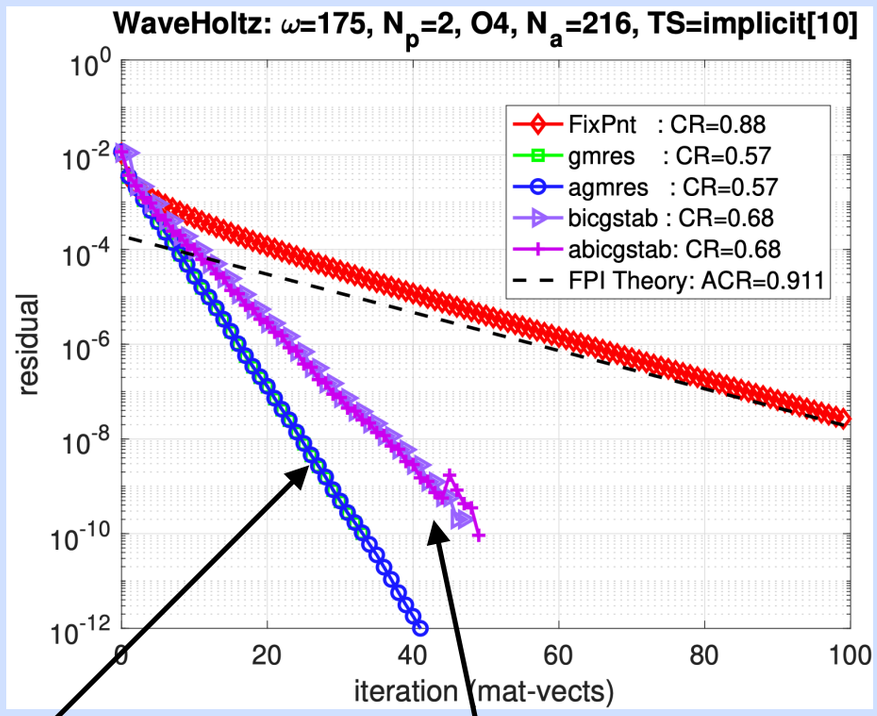




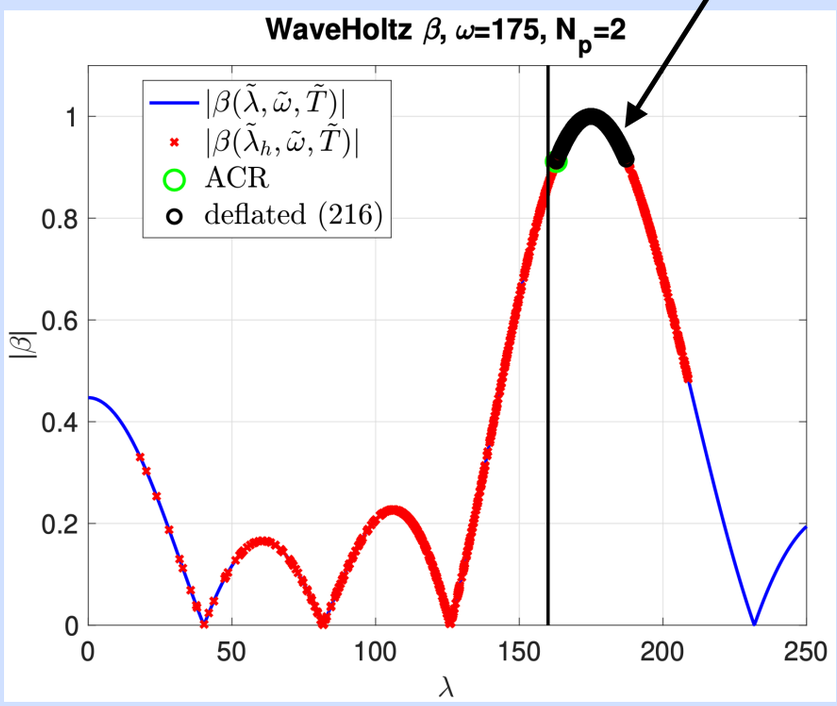
Convergence - no deflation

Convergence - with 64 deflated eigenmodes

Comparison: Direct Deflation versus Augmented Krylov Solvers

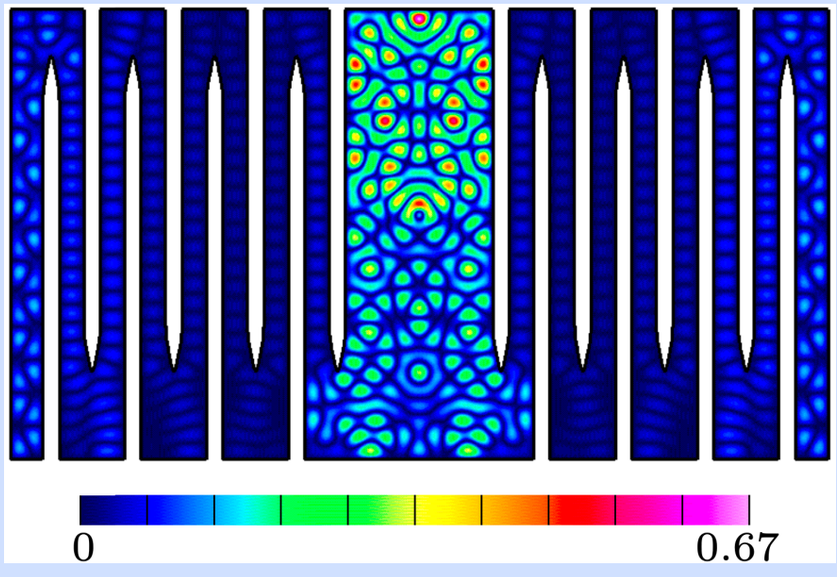


Deflated eigenvalues



GMRES + direct deflation matches Augmented GMRES

bi-CG-stab + direct deflation matches augmented bi-CG-stab

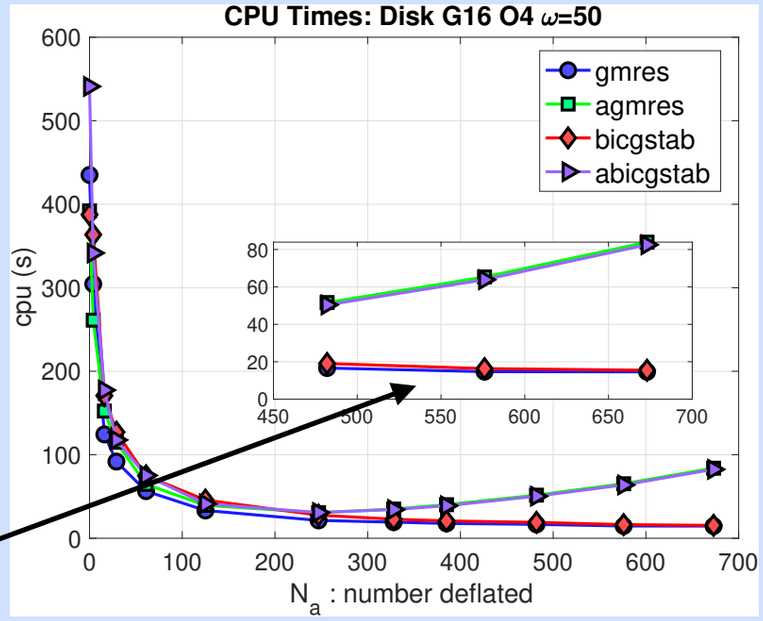


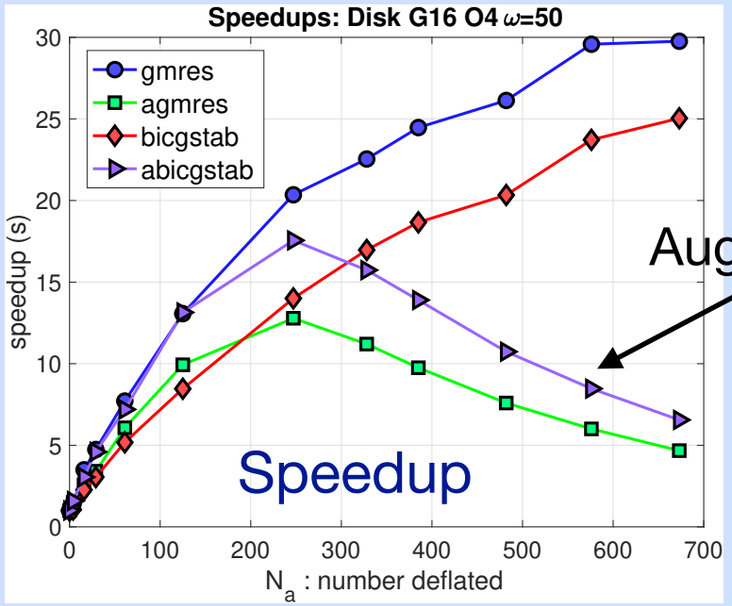
Convergence of Direct and Augmented Deflation is nearly identical

Eigenvector deflation - CPU times and speedups versus number of deflated eigenpairs (for a 2D disk)

CPU and speedup including startup costs

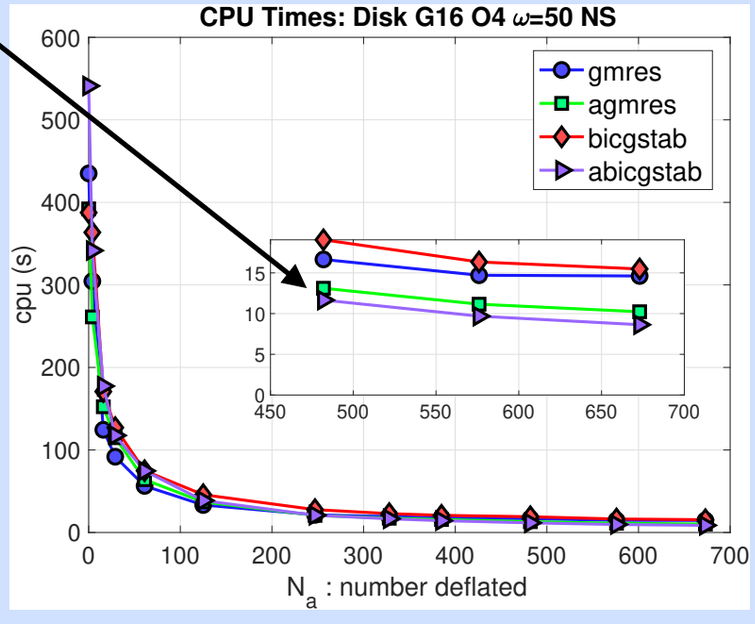
N_a = number of deflated eigenpairs

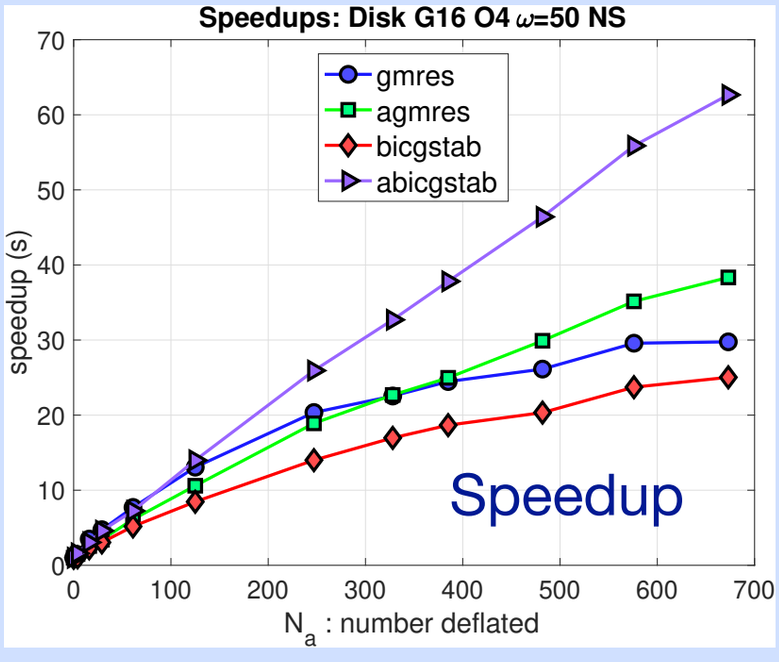




bi-CG-Stab beats GMRES for large N_a

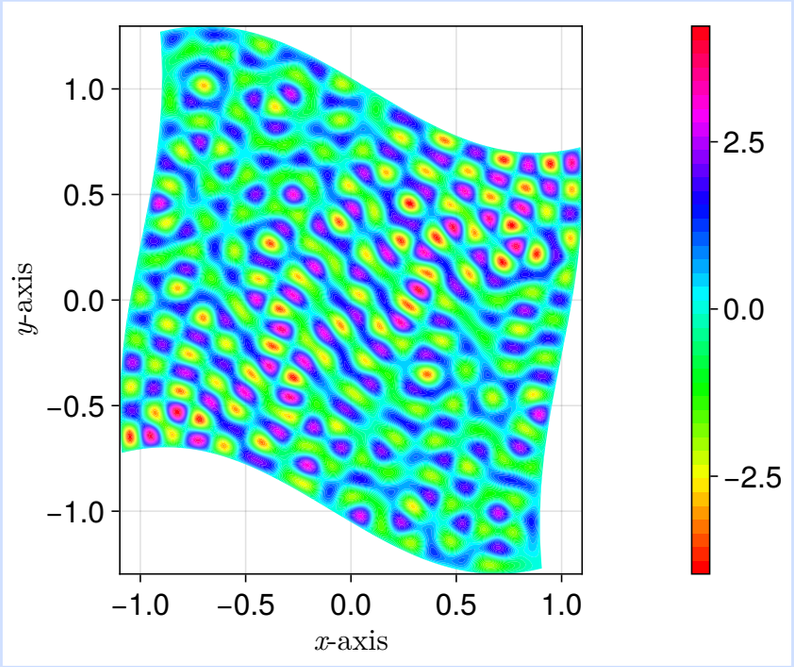
CPU and speedup excluding startup costs





Eigenvector Deflation - break even when including eigenvector computation

$\omega = 40$



Single grid example, Conjugate Gradient (CG) versus deflated CG

Cost to compute 100 eigenpairs = 593 wave solves
Cost to solve N_f deflated problems = $40\ N_f$ wave solves
Cost to solve N_f un-deflated problems = $500\ N_f$

N_f = number of right-hand-sides

Including the cost of computing the eigenpairs (using EigenWave) the deflated algorithm is faster when used to solve 2 or more RHS

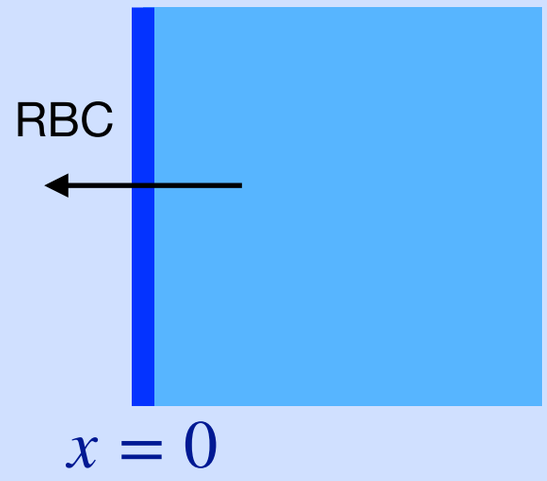
Open domains with radiation boundary conditions

Two radiation boundary conditions are considered in this talk

For a boundary at x=0

"Sommerfeld" (impedance):

$$w_t - c w_x = 0$$



Second-order Engquist Majda (EM):

$$w_{xt} - c(w_{xx} + \frac{1}{2} w_{yy}) = 0$$

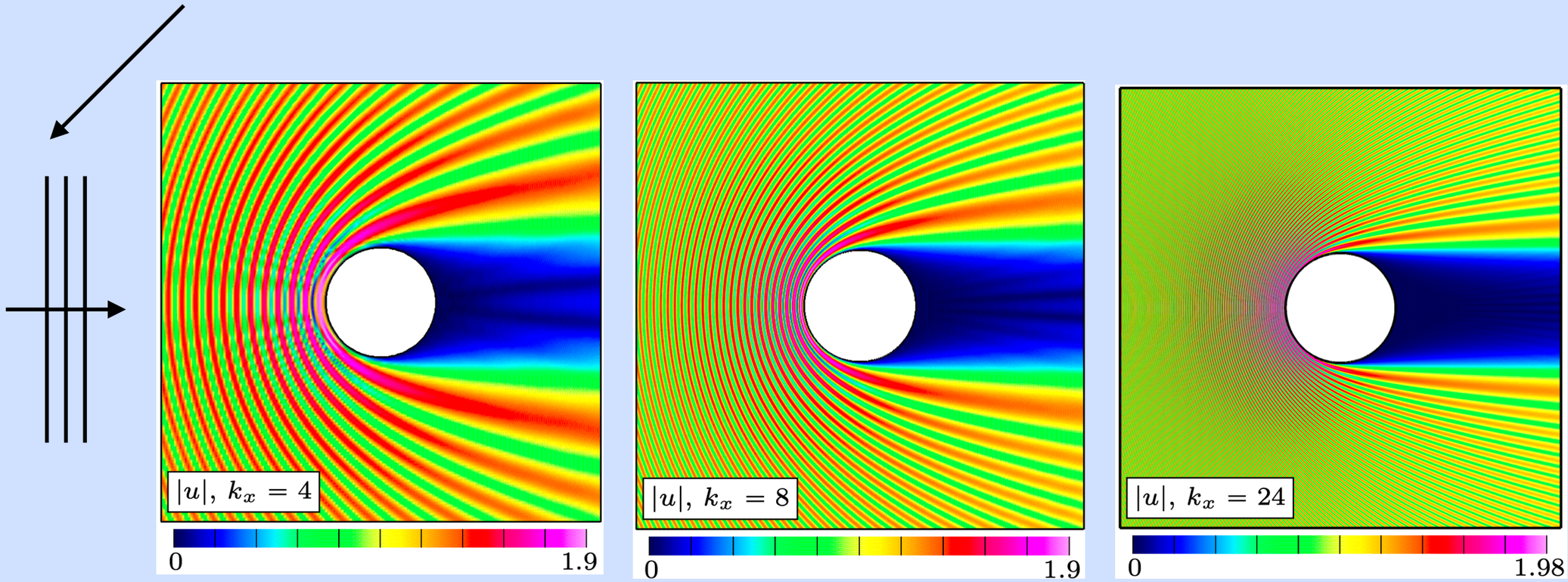
Some care is required to discretize these in a stable way

$$D_{+t} D_{0x} W_{\mathbf{j}}^n - \tilde{c} \left(D_{+x} D_{-x} + \frac{1}{2} D_{+y} D_{-y} \right) \left(\frac{1}{2} W_{\mathbf{j}}^{n+1} + \frac{1}{2} W_{\mathbf{j}}^n \right) = 0.$$

EM RBC, second-order accurate approximation,
For explicit or implicit time-stepping.

Example: WaveHoltz for high frequency scattering from a cylinder

Incident field: $w^I = a^I e^{i(k_x x - \omega t)}$ (we solve for the scattered field)



Absolute value of the (complex) total field.

Engquist Majda radiation boundary conditions are used on the outer boundaries.

The complex WaveHoltz algorithm iterates on $v^{(k)}(\mathbf{x})$ and $\dot{v}^{(k)}(\mathbf{x})$:

Algorithm 1 WaveHoltz Fixed-Point Iteration.

function $[u]$ = WaveHoltz$(\omega,\, f)$

$v^{(0)} = 0; \quad \dot{v}^{(0)} = 0;$ ▷ Initial guess

for $k = 0, 1, \ldots$ **do** ▷ WaveHoltz iterations.

$w(x,0) = v_{\mathbf{i}}^{(k)},$

$\partial_t w(x,0) = \dot{v}_{\mathbf{i}}^{(k)}$

$w(\mathbf{x}, [0, \bar{T}]) =$ solveWaveEquation$(\, v^{(k)},\, \dot{v}^{(k)},\, f\cos(\omega t)\,)$

$$v^{(k+1)} = \frac{2}{\bar{T}} \int_0^{\bar{T}} \left(\cos(\omega t) - \frac{1}{4}\right) w(\mathbf{x}, t)\, dt, \qquad \triangleright\ v^{(k+1)} = \text{Filter } w(\mathbf{x}, t).$$

$$\dot{v}^{(k+1)} = \frac{2}{\bar{T}} \int_0^{\bar{T}} \left(\cos(\omega t) - \frac{1}{4}\right) \partial_t w(\mathbf{x}, t)\, dt, \qquad \triangleright\ \dot{v}^{(k+1)} = \text{Filter } \partial_t w(\mathbf{x}, t).$$

end for

$u = v^{(k+1)} + (i/\omega)\, \dot{v}^{(k+1)}$ ▷ Complex Helmholtz solution $u(\mathbf{x})$

end function

The Complex WaveHoltz Algorithm defines a fixed point iteration:

When discretized, WaveHoltz defines a fixed-point iteration (FPI):

$$\begin{bmatrix} v_h^{(k+1)} \\ \dot{v}_h^{(k+1)} \end{bmatrix} = S_h \begin{bmatrix} v_h^{(k)} \\ \dot{v}_h^{(k)} \end{bmatrix} + \mathbf{b}_h$$

With solution satisfying a (real valued) linear system $A_h \mathbf{v}_h = \mathbf{b}_h$:

$$A_h \begin{bmatrix} v_h^{(*)} \\ \dot{v}_h^{(*)} \end{bmatrix} = (I - S_h) \begin{bmatrix} v_h^{(*)} \\ \dot{v}_h^{(*)} \end{bmatrix} = \mathbf{b}_h$$

Key point: This linear system is easier to solve with iterative methods

The basic FPI can be accelerated with matrix-free Krylov space methods such as GMRES.

Convergence of the WaveHoltz fixed-point iteration:

The asymptotic convergence rate (ACR) of the WaveHoltz FPI is

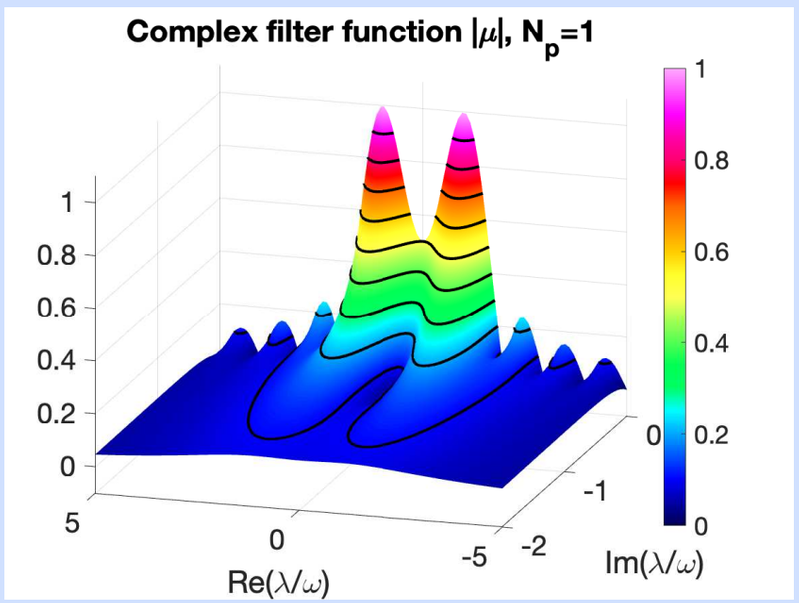
$$\text{ACR} = \mu^* = \max_{\lambda_h \in \Lambda_h} |\mu(\lambda_h, \omega)|$$

Max over eigenvalues

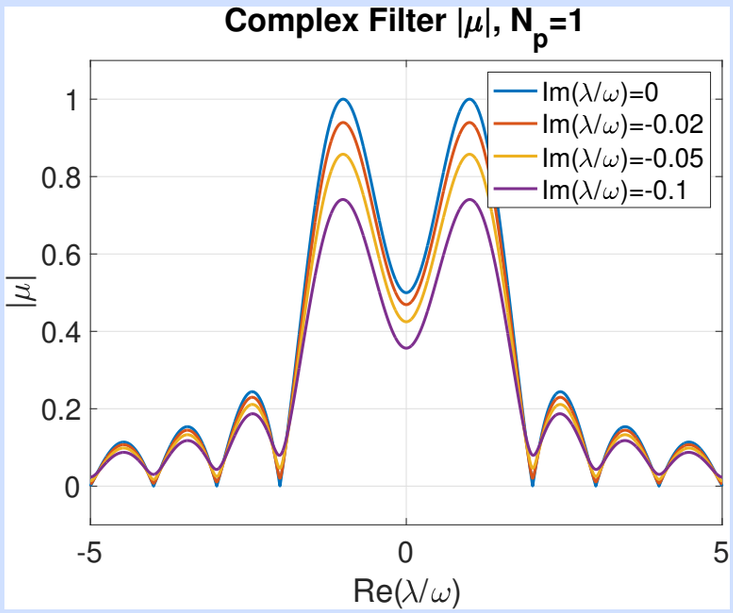
WaveHoltz complex filter function

$$\mu(\lambda;\omega) \stackrel{\text{def}}{=} \frac{2}{T_f} \int_0^{T_f} (\cos(\omega t) - \frac{1}{4})\, e^{-i\lambda t} dt \qquad T_f = \frac{2\pi N_p}{\omega}$$

N_p = number of periods over which the wave equation is solved



Surface plot of $|\mu|$

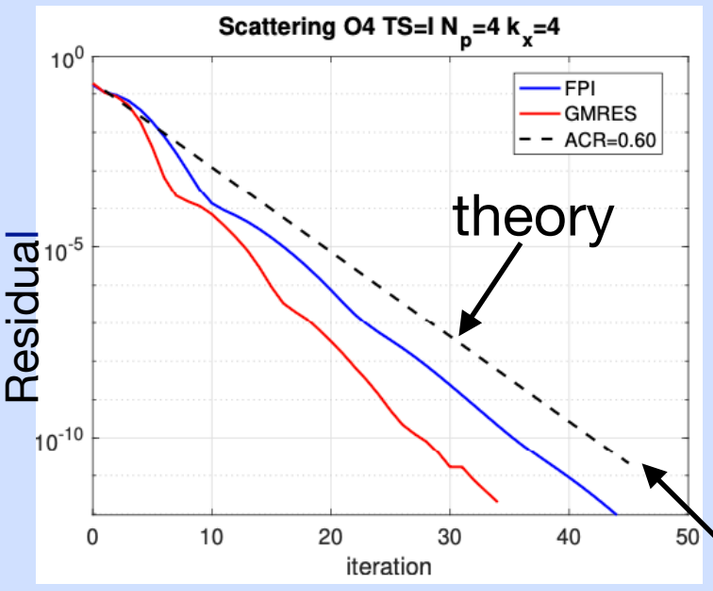


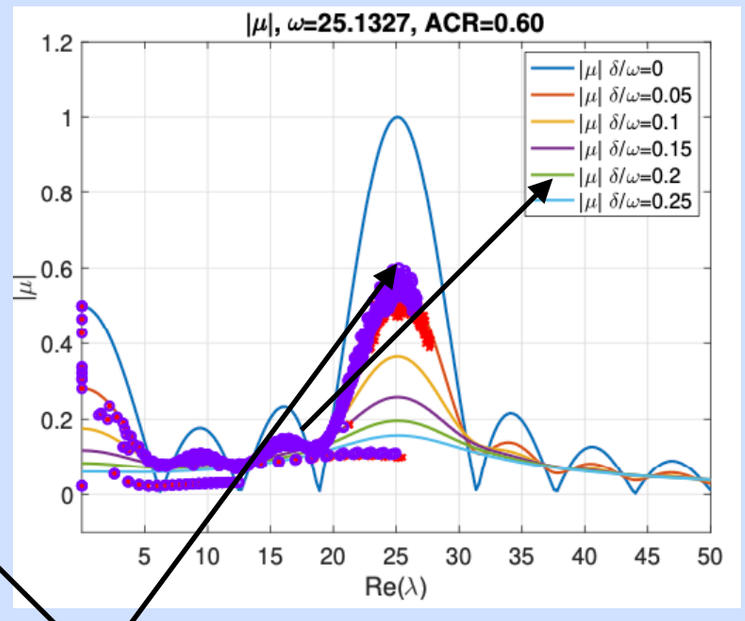
Cuts through surface plot of $|\mu|$

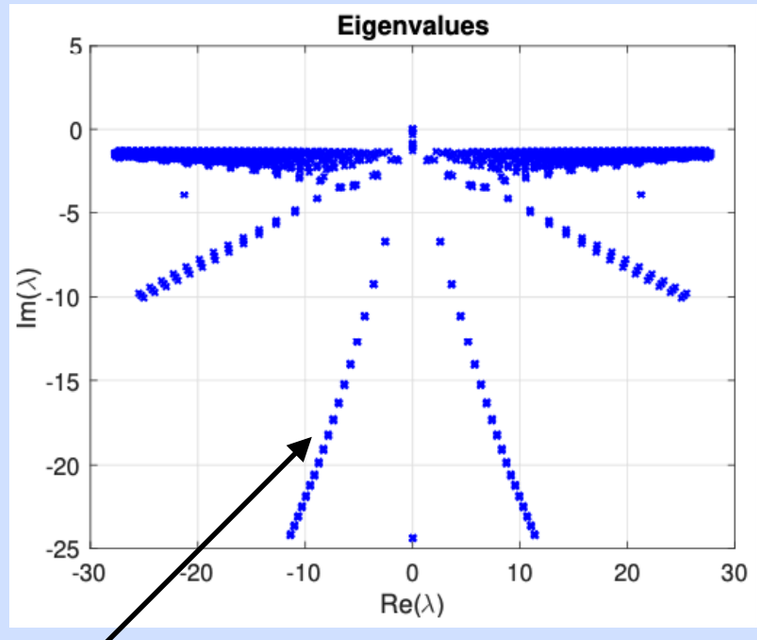
WaveHoltz convergence for scattering from a cylinder

The radiation boundary conditions add damping (energy loss) to the problem which leads to faster convergence rates of the WaveHoltz algorithm

Convergence curve with no damping







Convergence

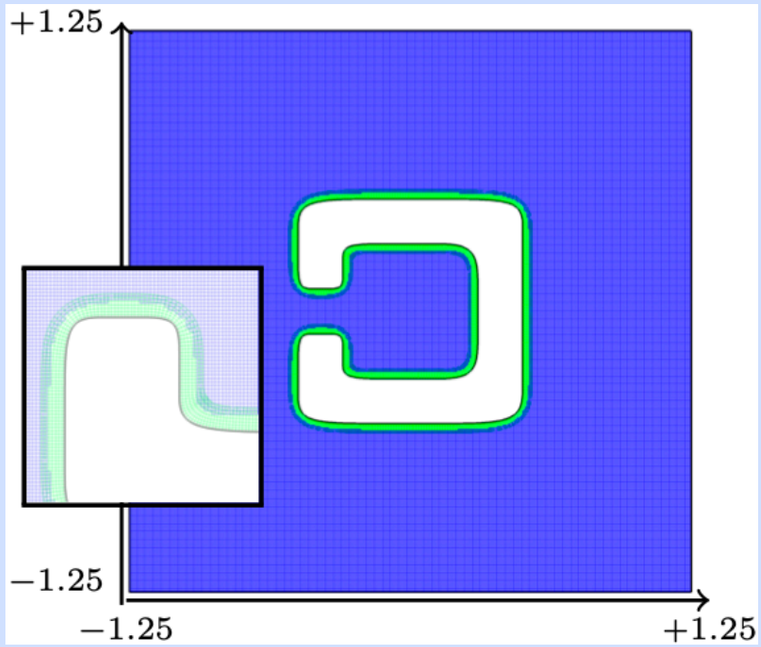
Estimated FPI convergence rate

Eigenvalues of Laplacian with Radiation BCs

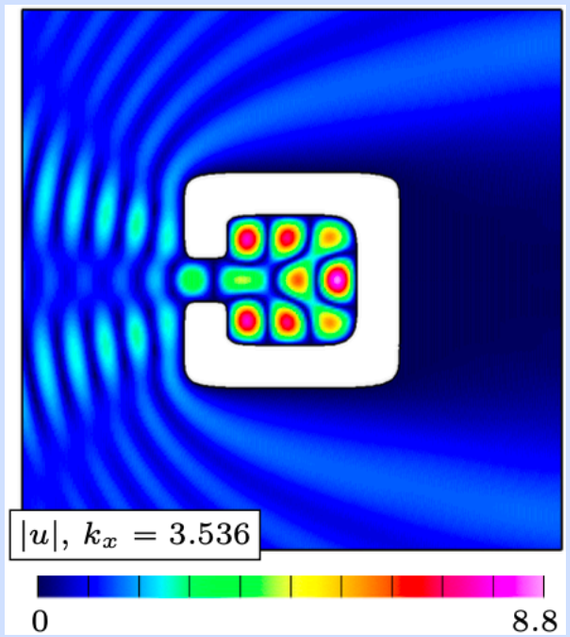
Negative imaginary in eigenvalue part implies damping

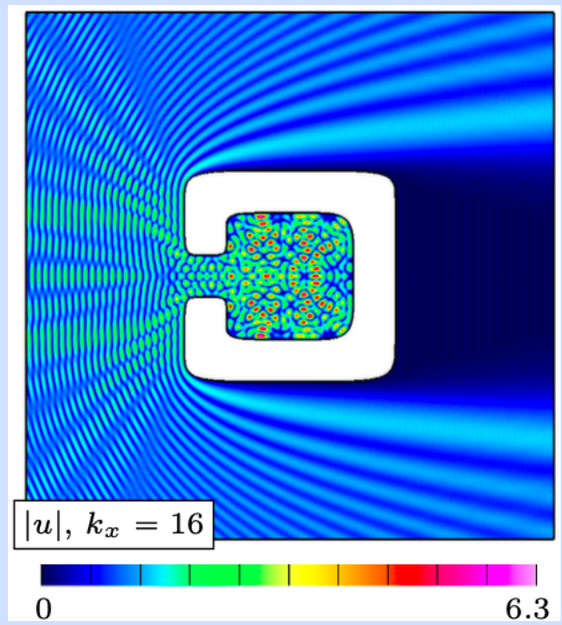
Scattering from a split ring resonator :

(a problem with nearly trapped modes)

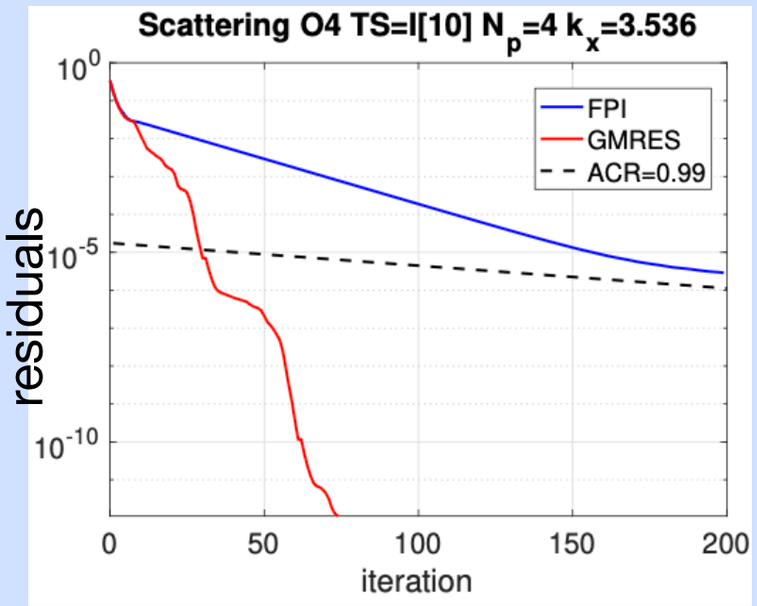


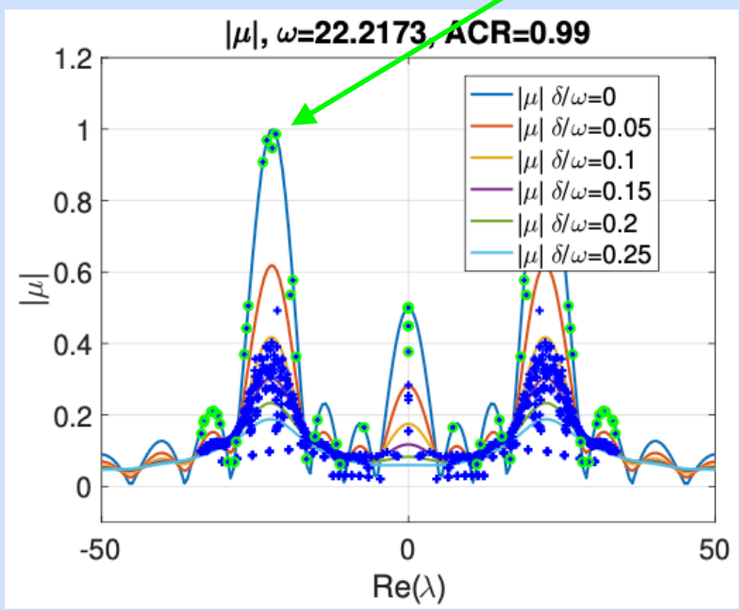
Overset grid

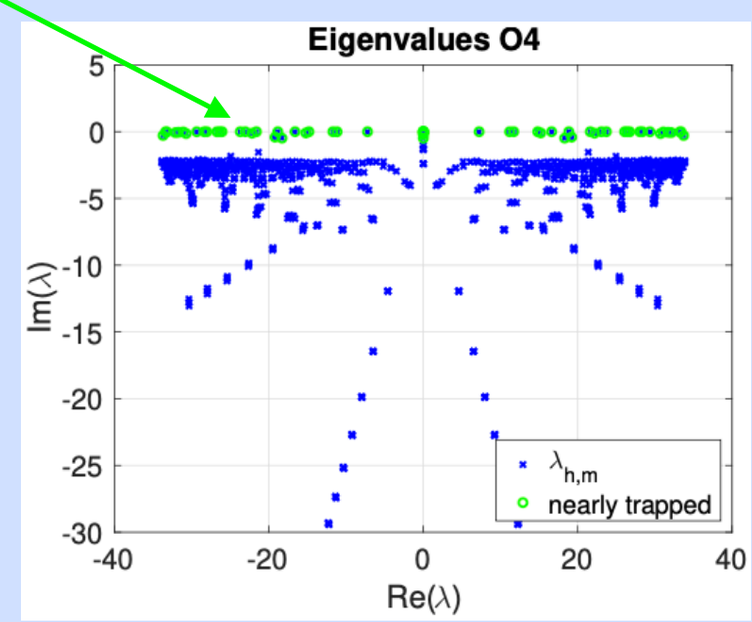




Cavity modes have little damping

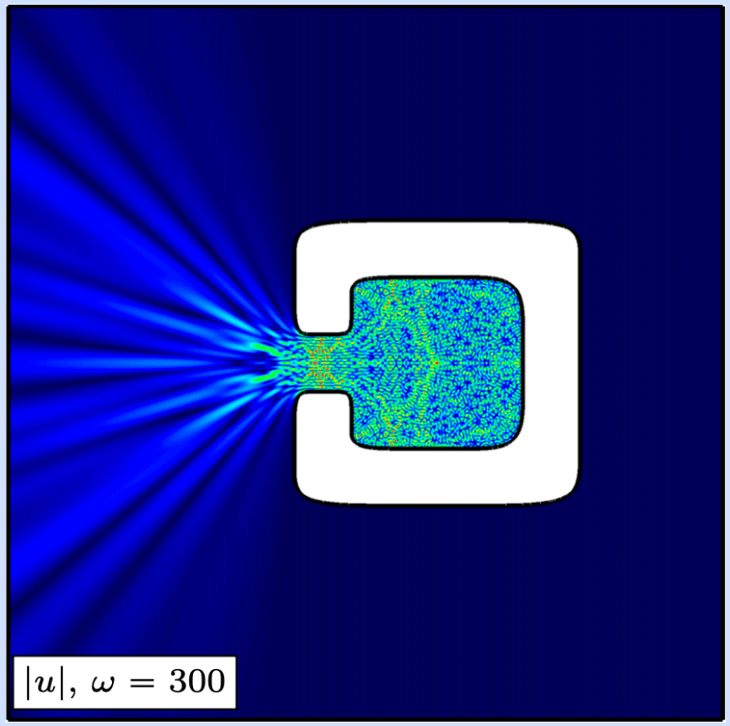




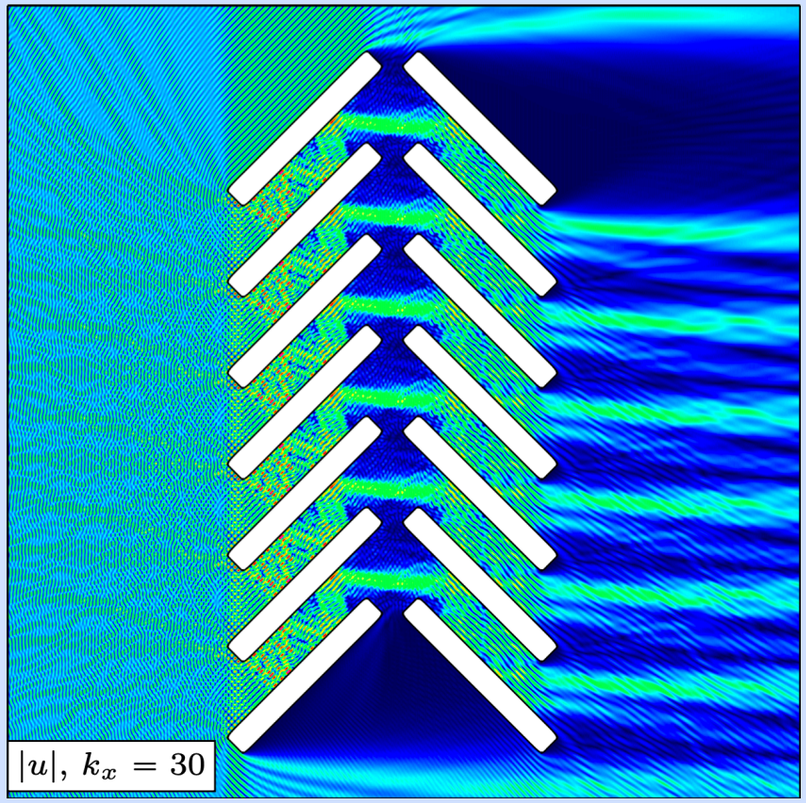


With nearly trapped modes, GMRES is much faster than the FPI

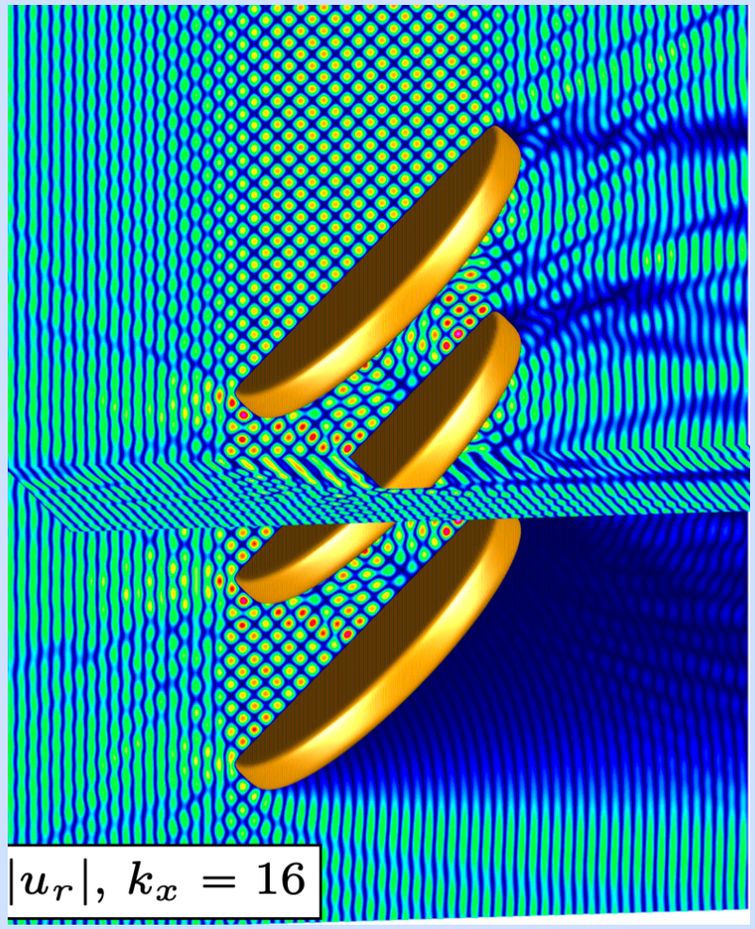
WaveHoltz can be easily parallelized (Here with MPI)



Gaussian source in the cavity



Scattering from multiple rectangles



Scattering from three disks in 3D

347 million grid points

End of the Talk